EV Technology Course — Module 10: E/E Architecture, Software and Connectivity

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A modern car may contain 70–150 electronic control units (ECUs), and the industry is collapsing them into a few central computers plus "zone" controllers. This shift, along with over-the-air (OTA) updates, is what people mean by the "software-defined vehicle": features, efficiency and even range can improve after the car is sold. EV makers have led this change because they started with a clean sheet.

1. Learning objectives

- Explain ECUs, in-vehicle networks (CAN, CAN FD, LIN, Ethernet) and gateways.
- Compare distributed, domain and zonal E/E architectures.
- Explain the software-defined vehicle, AUTOSAR and service-oriented architecture.
- Explain OTA updates and the cybersecurity rules that govern them (UNECE R155/R156, ISO/SAE 21434).
- Describe telematics and connected-car functions.
- Calculate bus load and OTA download time.

2. Fundamentals

- An **ECU** is a small computer that controls one function (e.g. windows, airbags, inverter).
- ECUs talk over **networks**: CAN is the classic automotive bus; Ethernet handles high data rates (cameras, central computers).
- Traditionally, each new feature added a new ECU from a supplier, each with its own software. Cars ended up with dozens of ECUs and kilometres of wiring.
- The new approach: a few powerful central computers run most of the software; **zone controllers** at the corners of the car connect local sensors and actuators and distribute power.

3. Detailed technical explanation

In-vehicle networks

Network	Speed	Use
LIN	~20 kbit/s	Simple sensors and actuators (window switches, mirrors)
CAN	125–500 kbit/s (up to 1 Mbit/s)	Powertrain, body, chassis messages; the EV workhorse (BMS, VCU, MCU)
CAN FD	Data phase up to ~5–8 Mbit/s	More data per frame; newer vehicles
FlexRay	10 Mbit/s	Deterministic chassis networks (declining)
Automotive Ethernet	100 Mbit/s – 10 Gbit/s	Backbone between zones and central computers; cameras; OTA

Architectures

	Distributed	Domain	Zonal + central compute
Organisation	One ECU per function	ECUs grouped by domain (powertrain, body, chassis, infotainment, ADAS)	Central computers run functions; zones handle I/O by location
ECU count	Highest	High	Lowest
Wiring	Longest	Long	Shortest
Software updates	Hard, ECU by ECU	Easier per domain	Centralised OTA
Example	Older vehicles	Many current ICE/EV platforms	Tesla, newer Chinese EVs, upcoming European platforms

Software-defined vehicle (SDV)

- Hardware is designed with spare capacity; software adds and improves features over time.
- **AUTOSAR Classic** for real-time, safety-critical ECUs; **AUTOSAR Adaptive** / Linux / hypervisors for high-performance computers.
- **Service-oriented architecture (SOA):** functions publish services ("vehicle speed", "open window") that other software can use, independent of where they run.

OTA updates

- **SOTA** (software over the air) and **FOTA** (firmware over the air) download through the telematics control unit (TCU), verify signatures, install to a backup memory bank, and switch over safely (often while parked).
- Regulations: **UNECE R156** (software update management) and **R155** (cybersecurity management system) apply in many markets; **ISO/SAE 21434** defines cybersecurity engineering.

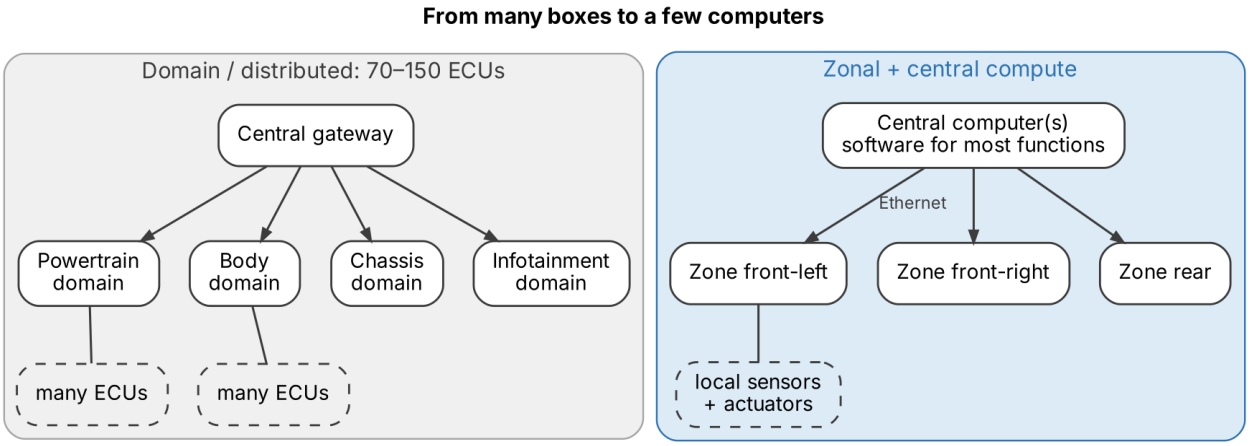
Cybersecurity basics

- Secure boot, hardware security modules (HSM), message authentication on CAN (SecOC), firewalls in gateways, intrusion detection, signed updates, secure key management.

Connectivity and data

- TCU with 4G/5G, GNSS and eSIM; app functions (remote AC, charging status, location); fleet telematics; battery health analytics in the cloud.
- India: AIS-140 requires vehicle tracking and emergency buttons on public transport and commercial vehicles; connected features are now common in Indian EVs.

4. Diagram: from many boxes to a few computers



5. Step-by-step: an OTA update that improves range

1. The OEM develops a new thermal-management calibration that cuts winter energy use.
2. It is validated (HIL, vehicle tests), signed with the OEM's private key and staged on the update server.
3. The car's TCU downloads the package over Wi-Fi or cellular.
4. The central computer verifies the signature and integrity.

5. The update installs into the inactive memory bank; the active software keeps running.
6. When parked and with enough 12 V and HV charge, the system switches banks and reboots the affected controllers.
7. If anything fails, it rolls back to the previous version; the result is reported to the OEM.

6. Engineering equations

Bus load:

$$\text{Bus load} = \frac{\text{bits per frame} \times \text{frames per second}}{\text{bit rate}}$$

Download time:

$$t = \frac{\text{size (bits)}}{\text{bandwidth (bit/s)}}$$

Raw camera data rate:

$$R = \text{pixels} \times \text{frame rate} \times \text{bits per pixel}$$

7. Worked examples

Example 1: CAN bus load. 1,500 frames/s of ~130 bits on a 500 kbit/s bus: 195,000 ÷ 500,000 = **39%**. Designers usually keep load below ~50–60% to protect timing.

Example 2: OTA download. A 2 GB package at 20 Mbit/s: 16,000 Mbit ÷ 20 = 800 s ≈ **13 minutes**.

Example 3: why cameras need Ethernet. An 8-megapixel camera at 30 frames/s and 12 bits/pixel produces about **2.9 Gbit/s** of raw data: thousands of times more than CAN can carry.

8. Real-world examples

- **Tesla:** a few central computers and body controllers; frequent OTA updates, including range and charging improvements.
- **Chinese EV makers (e.g. Xpeng, Nio, Li Auto, BYD):** central computing platforms and frequent OTA updates.
- **Volkswagen:** early software delays on the ID family led to the creation of its software unit and partnerships (e.g. with Rivian) to move to zonal architecture.
- **India:** Tata, Mahindra, MG and Hyundai offer connected-car apps and OTA updates on recent EVs; scope varies by model.

9. Manufacturing applications

- **End-of-line flashing:** each vehicle gets the right software set and configuration for its variant; flashing time is a takt-time issue.
- **Key provisioning:** cryptographic keys injected securely on the line.
- **Harness and connector reduction** with zonal architectures cuts assembly time and errors.
- **Traceability:** software versions linked to VIN for recalls and OTA campaigns.
- **Costing:** fewer ECUs and less copper, but expensive high-performance chips and large software teams; software becomes a major cost and value line.

10. Common mistakes and misconceptions

- **"OTA can fix anything."** It cannot fix hardware faults; some updates require homologation re-approval.
- **"Zonal means fewer wires only."** It also changes power distribution, safety design and supplier roles.
- **"CAN is obsolete."** CAN and CAN FD remain everywhere for control signals.
- **"Cybersecurity is an IT issue."** It is a vehicle safety and regulatory requirement.

11. Practical case study: a delayed software platform

Problem. An OEM's new EV is ready mechanically but launch slips by a year because software integration across 60 supplier ECUs fails validation.

Root causes. Each supplier delivered black-box software; interfaces changed late; no central integration environment; OTA not designed in from the start.

Response. Bring vehicle-level software in-house, consolidate functions into domain or central computers, adopt continuous integration with HIL farms, and design OTA from day one.

Lesson. In EVs, software capability is now a schedule and cost driver as big as hardware.

12. Revision notes

- ECUs talk over LIN, CAN, CAN FD and Ethernet.
- Distributed → domain → zonal + central compute.
- SDV: features via software; OTA via TCU with signed packages and rollback.
- R155 / R156 / ISO 21434 govern cybersecurity and updates.

- Bus load = bits × frames ÷ bit rate.

13. Glossary

Term	Meaning
ECU	Electronic control unit
E/E	Electrical/electronic architecture
Zone controller	Local hub for I/O and power by location
SDV	Software-defined vehicle
OTA / SOTA / FOTA	Over-the-air (software/firmware) update
TCU	Telematics control unit
AUTOSAR	Standard automotive software architecture
HSM	Hardware security module

14. Knowledge check

Conceptual

1. Why do cars end up with so many ECUs?
2. What are the advantages of zonal architecture?
3. Why does OTA need a backup memory bank?
4. Why is cybersecurity regulated for vehicles?
5. What stays on CAN even in zonal cars?

Numerical

1. CAN FD at 2 Mbit/s carries 600 frames/s of 600 bits. Bus load?
2. Download time of 500 MB at 10 Mbit/s?
3. A car moves from 100 ECUs to 3 central computers + 10 zone controllers. Percentage reduction?

Application

1. As a costing engineer, list the cost lines that rise and fall when moving to zonal architecture.
2. Draft an OTA safety checklist for a powertrain software update.

15. Answer key

Conceptual

1. Each new feature was traditionally sourced as a separate box with its own supplier software.
2. Shorter harness and less mass, fewer ECUs, central software and OTA, easier feature addition, better data access.
3. To keep a known-good version so the car always has working software if an install fails, and to install while still running.
4. Remote attacks could affect vehicle control and safety, privacy and fleets at scale.
5. Many real-time control signals and legacy components: powertrain, BMS, chassis and body actuators.

Numerical

1. 600 × 600 ÷ 2,000,000 = **18%**.
2. 4,000 Mbit ÷ 10 = **400 s** (about 6.7 minutes).
3. (100 − 13) ÷ 100 = **87%** fewer controllers.

Application

1. Fall: number of ECUs and housings, copper harness and connectors, assembly labour, supplier software integration. Rise: high-performance SoCs and memory, Ethernet switches, in-house software teams and tools, cybersecurity infrastructure, validation (HIL), cloud and OTA operations.
2. Signed package verified; compatibility with hardware variant; vehicle parked, gear in P, sufficient 12 V and HV SOC; inactive-bank install; post-install self-test; rollback path; regulatory relevance check (R156, homologation); staged rollout with monitoring.

16. Further learning

- AUTOSAR consortium public documents (autosar.org).
- UNECE R155 and R156 texts; ISO/SAE 21434.
- McKinsey and Roland Berger reports on software-defined vehicles.
- Vector Informatik e-learning on CAN, CAN FD and automotive Ethernet (free).

Next: Module 11, energy efficiency, range and consumption.